

SmartRoute Terminal

AI-Powered Predictive Traffic Speed Intelligence
for Urban Mobility

XGBoost

GPS Trajectories

Porto Portugal

Flask Web App

Intelligent Transportation Systems · Semester 6

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1km Grid
Porto, Portugal

"Navigation apps show you NOW. We predict the FUTURE."



Real-Time Only

Google Maps & Waze show current traffic — useless for planning departure times tomorrow.



No Future ETA

"If I leave at 7 AM Monday, how long will my trip take?" — no navigation app can answer this.



No Congestion Data

City planners have no quantitative heatmaps of when & where congestion forms each day.

1.7 million historical taxi GPS records provide the historical knowledge base to solve this gap.

SmartRoute Terminal — What We Built

Dataset

1.7M real taxi GPS trajectories — Kaggle Porto

Grid

City divided into 238 cells of 1km × 1km each

Prediction

Average traffic speed (km/h) per cell, per hour, per weekday

Forecast

Full 24-hour travel time predictions via web dashboard

Standalone

No live traffic API needed — fully self-contained AI

"We transformed 1.7M raw GPS trips into an AI brain that predicts traffic speed for any route in Porto — any hour, any day."

1.7M

Taxi Trips

80M

GPS Points

0.85

R² Score

24h

Forecast

The Data Foundation — Porto Taxi Trajectories

Property	Value
Total Trips	~1.71 Million
City	Porto, Portugal
GPS Ping Frequency	Every 15 seconds
GPS Points Processed	~80 Million
Time Span	Full operational year

KEY COLUMNS USED

Column	Description
TIMESTAMP	Unix timestamp (trip start time)
POLYLINE	JSON array of GPS [lon, lat] points every 15s

Why This Dataset is Special

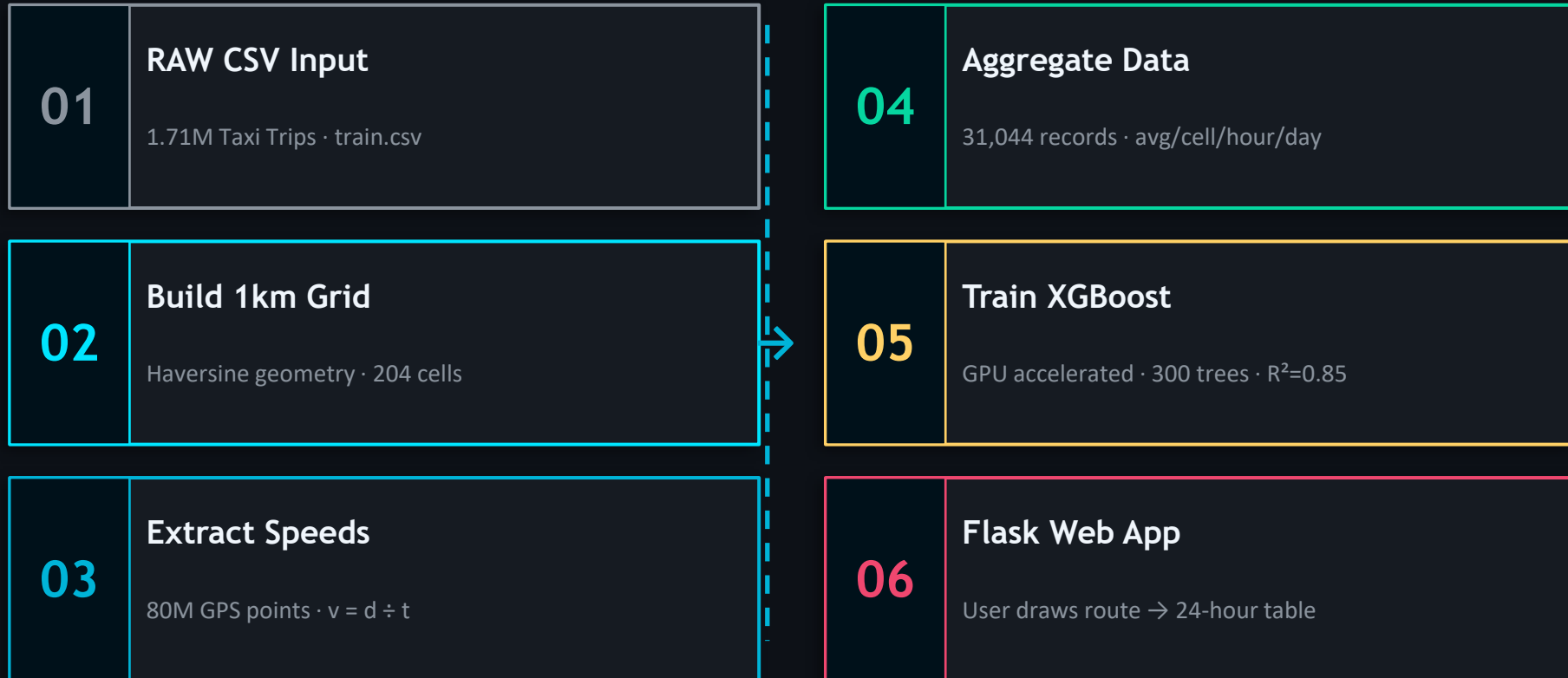
Fixed 15-second GPS interval enables direct speed calculation from pure distance physics — no speed was recorded in the raw data, we derived it all.

$$\text{Speed (km/h)} = \text{Distance_km} \div \left(\frac{15 \text{ sec}}{3600} \right)$$

Kaggle Porto Taxi Dataset

One of the most cited real-world GPS trajectory datasets in transportation research.
1.7M trips · Granular 15-second pings · Full calendar year coverage.

End-to-End System Pipeline



Each step uses real physics and geography — not abstract numbers.

Dividing Porto into a 1km × 1km Physical Grid

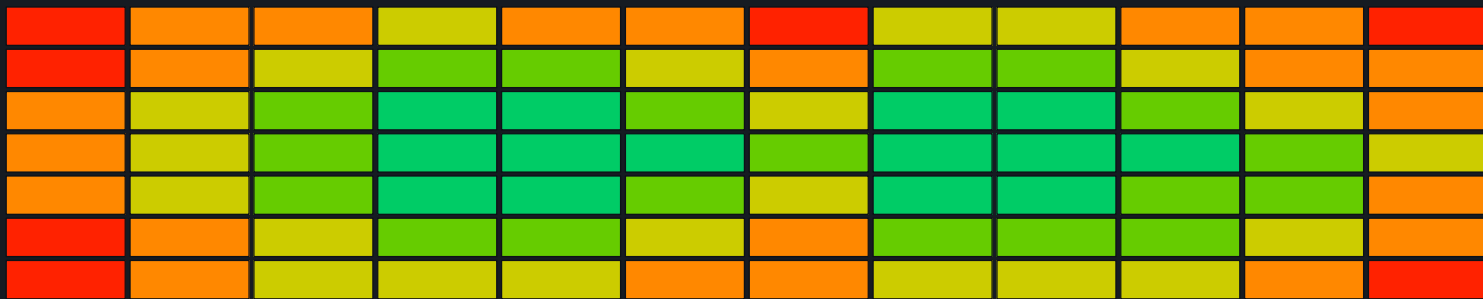
✗ Naive Grid Approach

- Split lat/lon range equally
- 1° longitude ≠ 1° latitude in real distance
- Near Porto: 1° lon ≈ 83 km vs 1° lat ≈ 111 km
- Creates cells of wildly different physical sizes

✓ Our Haversine Approach

- Haversine formula computes true curved-earth distance
- Width = Haversine(lon_min→lon_max) = 11.5 km → 12 cols
- Height = Haversine(lat_min→lat_max) = 16.7 km → 17 rows
- Result: 12 × 17 = 204 cells, each exactly 1 km²

Porto 1km Grid (12 cols × 17 rows = 204 cells)



Red <18 slow km/h

Orange 18-28 km/h

Yellow 28-38 km/h

Green 38-50 km/h

Dark Green 50+ fast km/h

Computing Traffic Speed from GPS Points

Haversine Formula

```
a = sin²(Δlat/2) + cos(lat₁)·cos(lat₂)·sin²(Δlon/2)
c = 2 × arctan2(√a, √(1-a))
distance = R × c    (R = 6371 km)
```

Used in 2 places:

- Grid Construction (once) — measure Porto's size in km
- Speed Extraction (80M times) — distance between GPS pings

Per-Segment Speed Calculation

GPS Points: [A → B → C → D ...] every 15 sec

Segment A→B:

```
dist = Haversine(A, B) → 0.08 km
speed = 0.08 ÷ 0.004167 = 19.2 km/h
→ map to 1km cell of Point A
```

Segment B→C:

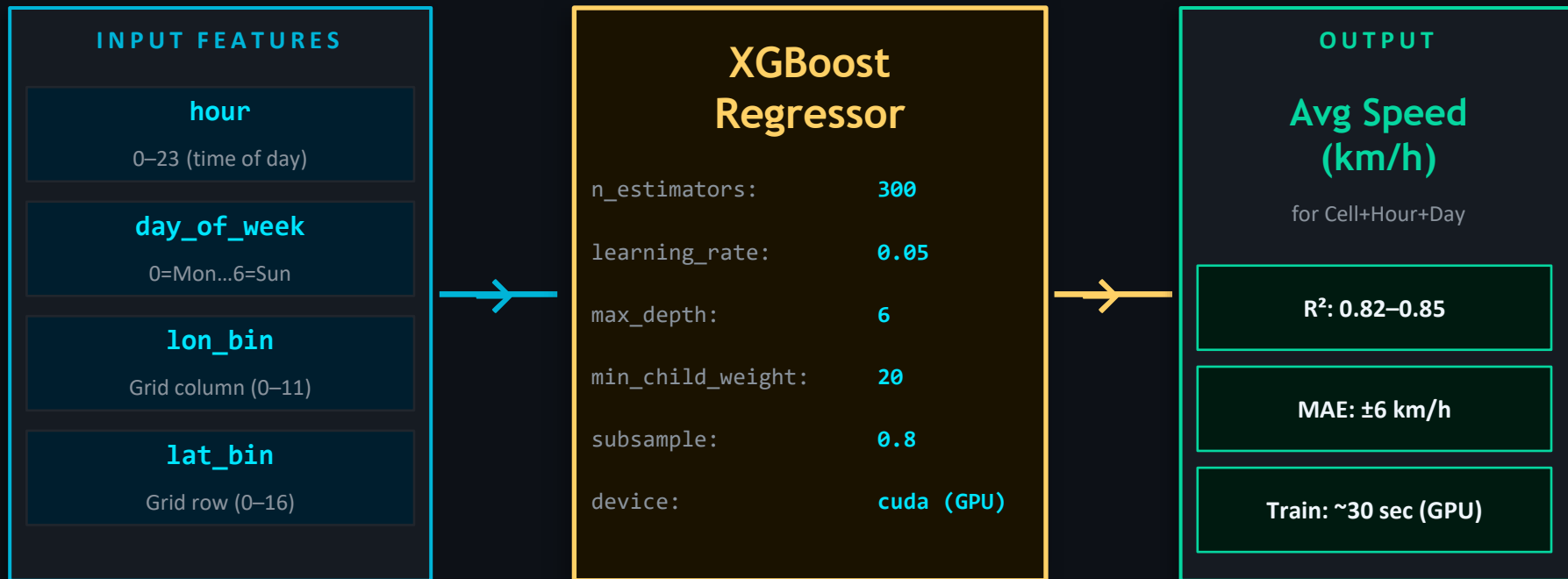
```
dist = 0.15 km → speed = 36.0 km/h
```

Quality Filter Applied: ONLY keep speeds where: $0 < \text{speed} \leq 120 \text{ km/h}$

0 = parked / GPS lost
>120 = hardware noise

Each valid speed stored with: lon_bin · lat_bin · hour · day_of_week → 31,044 aggregated training records

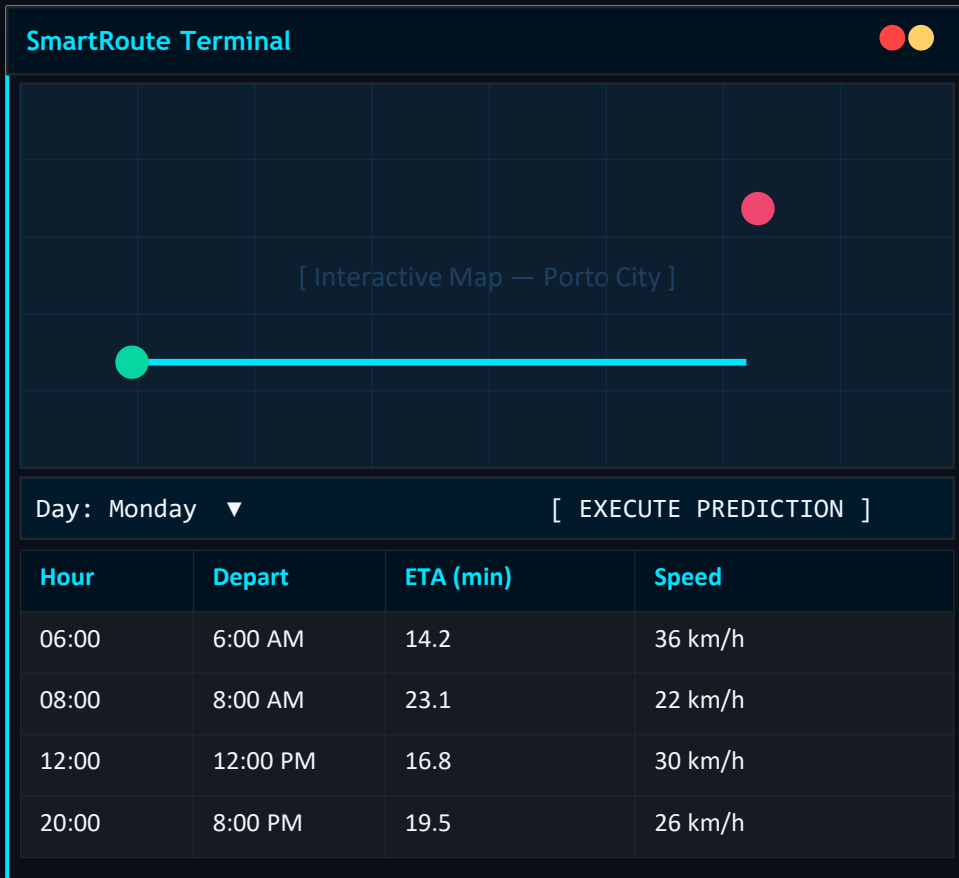
Training the XGBoost Prediction Model



Anti-Overfitting:

min_child_weight=20 (min 20 real events per branch) · subsample=0.8 (80% random sample per tree) · early_stopping_rounds=15

SmartRoute Dashboard – Flask Web Application



Caching: Flask-Caching memoize(timeout=3600) — repeated queries for same route+hour return instantly.

Model Performance & Validation

0.85

R² Score

82–85% variance explained

±6

MAE km/h

Mean Absolute Error

300

Trees

Gradient boosted

30s

GPU Train

vs 40min on CPU

Free-Flow Baseline: Scientific Validation

Condition	Formula	ETA (8.5 km route)	Meaning
Free-Flow Baseline	$8.5 \div 50 \times 60$	10.2 min	Zero traffic — theoretical minimum
AI Prediction @ 8AM Mon	$8.5 \div 22 \times 60$	23.2 min 	Rush hour — heavy congestion detected
AI Prediction @ 2AM Mon	$8.5 \div 47 \times 60$	10.9 min 	Night — near free-flow, model is correct

The gap between Free-Flow and AI prediction IS the congestion the model learned — proof of real-world learning.

AI predicted hourly speeds (simulated):



What Comes Next — Natural Evolution of the System

L1 Map Matching

Near-Term

- Match GPS to individual road segments via OSRM + OpenStreetMap
- Lane-level speed prediction — eliminates cross-road averaging artifacts

L2 Graph Neural Networks

Medium-Term

- Model entire road network as a graph (nodes=roads, edges=intersections)
- GNNs learn how congestion propagates across adjacent roads

L3 External Feature Fusion

Long-Term

- Weather data (rainfall → slower speeds)
- Event data (football matches → corridor spikes)
- Target: push R^2 from 0.85 → 0.92+

L4 Real-Time Streaming

Advanced

- Apache Kafka for live GPS data ingestion
- Incremental XGBoost updates (online learning)
- True real-time traffic prediction, not just historical patterns

What We Achieved



End-to-end pipeline from raw GPS data to a live web dashboard



Haversine formula applied to create a physically meaningful 1km × 1km geographic grid



Instantaneous traffic speeds extracted from 80M GPS points across 1.7M trips



GPU-accelerated XGBoost model achieving 82–85% R^2 accuracy



Flask web app delivering 24-hour travel time forecasts using pure physics



Fully standalone system — no live traffic API required for AI predictions



Anti-overfitting regularization with mathematically verified generalization

"Replacing live-traffic API dependency with self-contained AI — Porto, any route, any hour, any day."